1/17/19

MAE 575: Lecture 2

Prop/turboprops:
- very fuel efficient due to high $\dot{m}$ & low V_e
- limited range
- Used in regional airlines

High BPR turbofans:
- Still fuel efficient due to high $\dot{m}$
- Capable of high subsonic/transonic flights.
- Used in commercial jets.

Low BPR turbojets:
- Capable of supersonic flight
- engines for F22; (F22 employs 'supercruise': can fly supersonic without afterburners)

Ramjets : very low M# in combustor ($M_c \sim 0.1$)

Scramjets : Mach # in combustor is supersonic ($M_c \approx 1.2 - 2.5$)

Pulse jet : subsonic deflagration, $M_c \sim 0.4$

PDE : detonation

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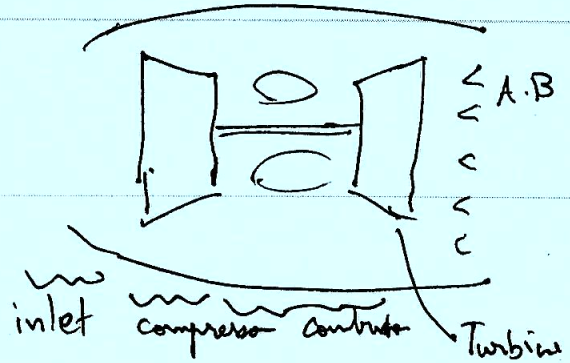
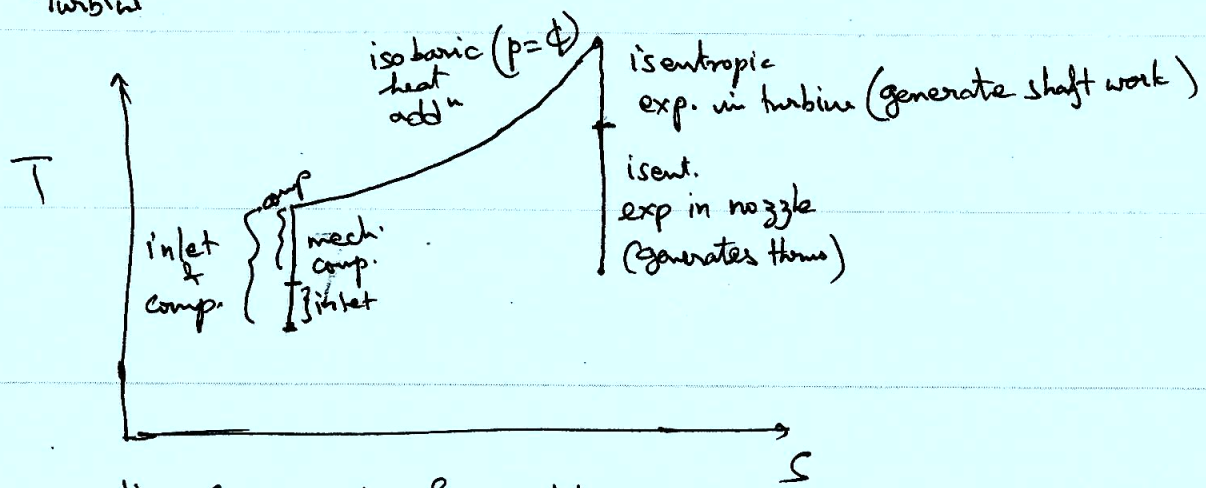
Turbojet (Brayton Cycle)





- As flight Mach increases, portion of compression from inlet compression increases, decreasing the need for mech. compressor
- Above M = 3, don't need compressor at all (Don't need turbines either)
- Ramjet looks the same as turbojet in T-S. space.

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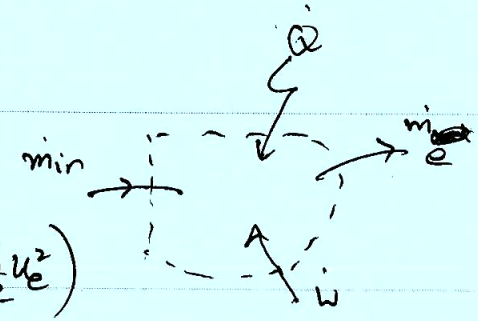
H-k diagrams

- relationship between stagnation & static enthalpy.
- From cons. of energy in CV, 1D flow.

$$h = e + pv$$



$$\dot{Q} + \sum \dot{m}_{in}\left(h_{in} + g z_{in} + \frac{1}{2} u_{in}^2\right) = \dot{W} + \sum \dot{m}_e \left(h_e + g z_e + \frac{1}{2} u_e^2\right)$$

For adiabatic flow, w/ no other forms of work (excluding flow work) & negligible pot. energy changes, 1st law reduces to

$$\sum \dot{m}_{in}\left(h_{in} + \frac{1}{2} u_{in}^2\right) = \sum \dot{m}_e \left(h_e + \frac{1}{2} u_e^2\right)$$

If we have only a single inlet & outlet, cons. of mass $\dot{m}_{in} = \dot{m}_e$

$$h_{in} + \frac{1}{2} V_{in}^2 = h_e + \frac{1}{2} V_e^2$$

@ Stag^n, $V_{stag^n} = 0$

$\Rightarrow$ Stag^n enthalpy : $h_0 = h_{in} + \frac{1}{2} V_{in}^2$

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Stag^n enthalpy = static enthalpy + KE

= enthalpy enhance if flow is isentropically decelerated to zero vel.

Recall heat capacity

$$C_p \equiv \left(\frac{\partial h}{\partial T}\right)_p$$

$$C_v \equiv \left(\frac{\partial e}{\partial T}\right)_v$$

$\forall = \text{¢} \Rightarrow$ all energy added to T increase

$p = \text{¢} \rightarrow$ some energy goes to work $(p\Delta\forall)$

For an ideal gas, $\left.\begin{array}{l} h = f(T) \\ e = f(T) \end{array}\right\}$ only (TPG)

& so

$$C_p = \left.\frac{dh}{dT}\right)_p \quad \text{for ideal gas}$$

$$\Delta h = \int dh = \int C_p(T)\,dT$$

For a calorically perfect gas. $C_p = \text{¢}$

$$\Delta h = C_p \Delta T$$

Let $h = 0$ @ $T = T^0 \rightarrow T^0 =$ ref temp.

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So, $\because$ $h = C_p T$ $\hookrightarrow$ T is relative to ref temp

So, for CPG

$$h = C_p T$$

$$h_0 = C_p T_0 \quad ; \quad T_0 : \text{stag}^n \text{ temp.}$$

1st Law : w/ previous assumption

$$h_0 = h + \frac{1}{2} V^2$$

$$C_p T_0 = C_p T + \frac{1}{2} V^2$$

$\div \; C_p T_0$

$$1 = \underbrace{\frac{C_p T}{C_p T_0}}_{\substack{\text{dimensionless} \\ \text{static enthalpy} \\ \equiv H}} + \underbrace{\frac{\frac{1}{2} V^2}{C_p T_0}}_{\substack{\text{dimensionless} \\ \text{K.E} \\ \equiv K}}$$

$$\underline{1} = H + K$$

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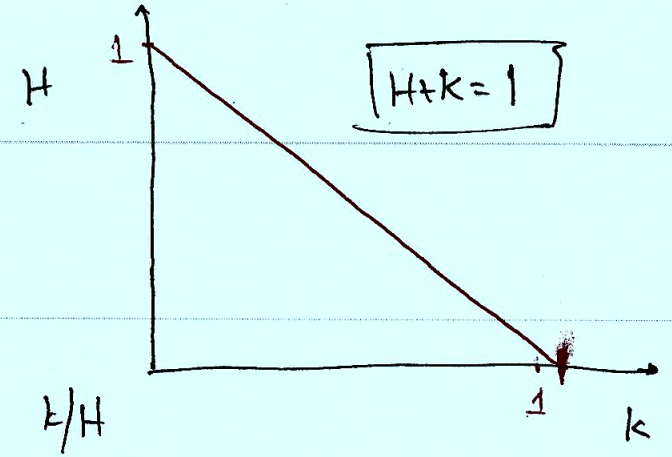
Thus, 1st law for isentropic process w/o work, involving a CPG in H-K space :



Look @ the ratios.

Vel.	Temp. (static)	Appl.	$K = \frac{\frac{1}{2}V^2}{C_pT_0}$	H	K/H
30 m/s	300K	Car.	0.001	0.999	1.5×10^{-3}
M = 1	223K	Com. airline	0.168	0.832	0.2
M = 8	223 K	Scramjet (JP8)	0.93	0.07	13
M = 12	223	Scramjet (H_2-powered)			30

⇒ Must change our mindset & look @ K.E as the primary component that contributes to h_0

$h_0 = ¢$ across a shock (adiabatic, but not reversible)

h_0 changes w/ heat add^n or ext. work (comp, turb.).

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→ Will use H-K diagrams to show ramjet / scramjet cycles, rather than T-s. space.

→ Because we are interested in raising the stag^n temp. thru heat add^n (combustion), which is (a Rayleigh flow problem), we will define H & K as being normalized by freestream stag. enthalpy

$$H = \frac{C_p T}{C_p T_{0,in}}$$

$$K = \frac{V^2}{2 C_p T_{0,in}}$$

And so for a propulsion system

$$H + K \geq 1 \qquad (=1 \text{ for adiabatic})$$

b/c of combustion changing T_0

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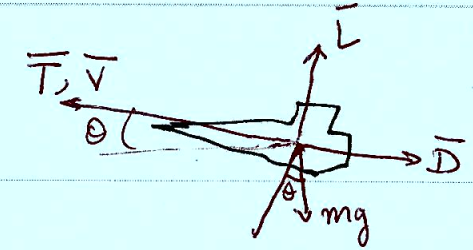
Ch 2.

Atmospheric flight



Consider a vehicle flying @ flight path angle θ; thrust & weight mg acts thru center of mass (COM) drag (D), lift (L) thru the center of pr. (COP). We will assume COP & COM are co-located

→ Lift is always normal to drag.

→ Drag is (almost) always ∥ to thrust.

→ Attach a coord system to vehicle, such that T, D are on one axis & L is the orthogonal component.

Apply Newton's 2nd law.

$$\sum \bar{F} = \frac{d}{dt}(m\bar{V}) = m\frac{d\bar{V}}{dt} + \bar{V}\cancel{\frac{dm}{dt}}^{\,0}$$

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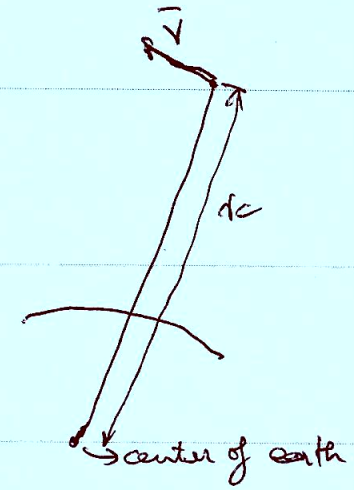
Along T, D axis

$$m\frac{d\bar{V}}{dt} = T - D - mg\sin\theta$$

(assume T & D are $\parallel^{el}$ to instantaneous vel $\bar{V}$)

Along L axis

$$m\frac{d\bar{V}}{dt} = mg\cos\theta - L$$



r_c : radius of earth at altitude

Centrifugal force

$$m\frac{dV}{dt} = m\frac{V^2}{r_c}$$

Thus normal to inst. $\bar{V}$.

$$\frac{mV^2}{r_c} = mg\cos\theta - L$$

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